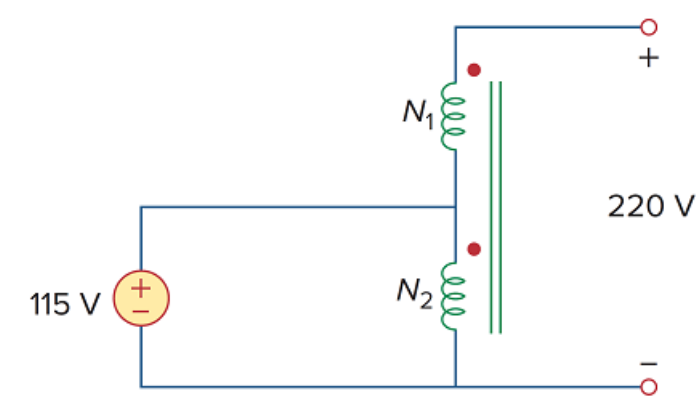


## Problem

When individuals travel, their electrical appliances need to have converters to match the voltages required by their appliances to the local voltage available to power their appliances. Today these converters use power electronics to convert voltages. In the past these converters were autotransformers. The autotransformer shown in Fig. 13.134 is used to convert 115 to 220 V. What is the value of the turns? If the maximum current available from the 115 V source is 15 A, what will be the maximum current available for the 220-V appliance?

Figure 13.134



## Step-by-step solution

## Step 1 of 3

Refer to Figure 13.134 from the textbook.

Observe that the given circuit is a step-up auto transformer.

Write the expression for the circuit voltage ratio,  $\frac{V_1}{V_2}$  for the given step-up auto transformer.

$$\frac{V_1}{V_2} = \frac{N_2}{N_1 + N_2} \dots\dots (1)$$

Here,

$V_1$  is the primary side voltage,

$V_2$  is the secondary side voltage.

## Step 2 of 3

Substitute 115 V for  $V_1$  and 220 V for  $V_2$  in equation (1).

$$\frac{115}{220} = \frac{N_2}{N_1 + N_2}$$

$$\frac{23}{44} = \frac{N_2}{N_1 + N_2}$$

$$23(N_1 + N_2) = 4N_2$$

$$23N_1 = 21N_2$$

$$\frac{N_1}{N_2} = \frac{21}{23}$$

$$\frac{N_1}{N_2} = 0.913$$

Therefore, the turns ratio is,  $\frac{N_1}{N_2}$  is 0.913.

### Step 3 of 3

Assume the maximum current available at 220 V side to be  $I_2$  for a maximum current of 15 A on the primary side.

Since power is same at both the sides of the autotransformer.

Thus,

$$V_1 I_1 = V_2 I_2$$

$$I_2 = \frac{V_1 I_1}{V_2}$$

Substitute 115 V for  $V_1$  and 220 V for  $V_2$  and 15 A for  $I_1$ .

$$\begin{aligned} I_2 &= \frac{(115)(15)}{(220)} \\ &= 7.841 \text{ A} \end{aligned}$$

Therefore, the maximum current available at 220 V side,  $I_2$  is 7.814 A.